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Controler Holguin

Practica 5 Diseño de controlador para un sistema de segundo orden

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] \quad (1)$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L \frac{d[I_1(s) - I_2(s)]}{ds} + R[I_1(s) - I_2(s)]$$

$$Ls[I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

Procedimiento algebraico

$$V_e(s) = (Ls + 2R)I_1(s) - (Ls + R)I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} \left(Ls + \frac{1}{Cs} + 3R \right) I_2(s) = \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{Cs} \right) I_2(s) = \frac{CRs + 1}{Cs} I_2(s)$$

Funcion de transferencia

$$\frac{V_s}{V_e} = \frac{\frac{CRs + 1}{Cs} I_2(s)}{\frac{3CLRs^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs} I_2(s)}$$

$$\frac{V_s}{V_e} = \frac{(CRs + 1)(Ls + R)}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_s}{V_e} = \frac{CLR s^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

Capacitor $470 \mu F$
 Resistencia ~~2.2k~~ $2.2k$
 Inductor ~~22mH~~ $22mH$

Error en el estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s V_e(s) \left(\frac{1}{s} - \frac{(L R s^2 + C R^2 + L) s + R}{3 C L R s^2 + (C S C R^2 + L) s + 2 R} \right)$$

$$e(s) = \frac{1}{2} \rightarrow 0.5$$

Estabilidad en lazo abierto

$$3 C L R s^2 + (C S C R^2 + L) s + 2 R = 0$$

$$R = 2.2k$$

$$C = 470 \mu F$$

$$L = 22mH$$

$$a = 3(2.2 \times 10^3)(470 \times 10^{-6})(22 \times 10^{-6})$$

$$b = [s(2.2 \times 10^3)^2(470 \times 10^{-6})] + 2 \times 2.2 \times 10^3$$

$$c = 2(2.2 \times 10^3)$$

El sistema es estable
 con un respuesta sobrepasada

Gráfica de la señal

